Novel Stochastic Computing using Amplitude and Frequency Pulse Encoding

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***Abstract*—Stochastic computing (SC) is a type of logic computation that converts binary numbers to stochastic bitstream. It is a pseudo-analog computation in the digital domain that can realize multiplication, addition and complex function computations. In this paper, we present a new encoding method AFE(Amplitude and Frequency Encoding) for SC, which extend SC to use multibit streams instead of bit streams to represent data. This new method still use the expectation of streams to achieve the computation. Compared with conventional stochastic bitstream, AFE SC proposed realize low-latency and low-area occupation instead of the long latency of classic method. And we also discuss the rationality of circuits from a mathematical perspective. The hardware logic circuits of AFE-SC including multiplier, adder and converter are designed and implemented by FPGA. In addition, the latency of computing, area and precision of AFE-SC is measured based on FPGA board.**

Keywords—Stochastic computing, Low latency, Amplitude and Frequency encoding

I.INTRODUCTION

Stochastic computing(SC) is one of the important methods to realize pulse computation[1], it was originally advocated by Gaines[2] in 1963. In the past of few years, SC becomes popular again due to its extremely low power consumption and small hardware occupation of computation core. And it has been applied to a wide variety of applications, such as image processing [3]-[6], neural networks [7],[8], and deep learning [9]-[11].

In SC paradigm, digital logical computation is performed on stochastic bitstreams instead of binary number. The value of stochastic bitstream is represented by the probability of firing “1” based on the stream length of accumulation time.

This particular numerical representation has a great advantage of the occupation of hardware in processing complex functions and polynomial operations[2]. Since data in circuits are usually recorded as binary representation, the first step of SC is that using BN (Binary Number) to SB (Stochastic bitstream) converter to get SC streams. But BN-SB converter will also increase consumption remarkably [12],[13]. Because of the problem of SC that high latency, poor precision and energy efficiency, some research groups have made some improvements on computing rules for conventional frequency encoding (FE) including bipolar or unipolar representation.

H. Sim et al.[10] propose a new computing rule to get better latency; Liu et al. [14-16] use Sobol to optimize precision; D. Jenson et al.[17-19] propose deterministic method to achieve accurate calculation. Parhi et al. [20] propose hybrid formats of unipolar and bipolar to reach better energy efficiency; V.T. Lee et al. [21] propose a digital analog hybrid method, which is more energy-efficient than binary logic method.

But there are some problems caused by encoding method instead of computing rules, such as high latency. The main problem of bit stream is its efficiency of carrying information, when stream length is equal to N, it can only represent N different data, which is weaker than binary representation (n-bit can represent 2^n different data).

Based on the above research results and remaining challenges, we propose new design ideas and methods. The main contributions of this paper are as follows:

- A new encoding method: AFE (Amplitude and Frequency Encoding) of SC is proposed for improving latency and area efficiency;
- The circuit of multiplier, adder, BN-SB and SB-BN converter for AFE is designed;
- The circuit of quadratic polynomial based on AFE SC is tested, which make latency decrease exponentially.

The remaining manuscript is organized as follow. Section II, existing SC methods are introduced, the new encoding method, AFE SC, is discussed and the circuit of AFE SC is designed. Section III, the error of AFE SC is discussed, the precision and area efficiency of conventional and AFE SC are compared. Section IV, the conclusion of this paper is presented.

II.STOCHASTIC COMPUTING

A.Existing Methods of SC

The classic SC[2] for a quadratic polynomial calculation is shown by Fig.1(a), BN-SB is composed of RNG(random number generator) and comparator. Through the RNG circuit, A and B are converted into stochastic stream S_A and S_B. The probability of two stream being "1" are $P(S_A)=A/2^n$ and $P(S_B)=B/2^m$. After passing through “AND” gate, a new stochastic stream $S_{A,B}$ is obtained, the probability of the output stream being "1" is $P(S_{A,B})=P(S_A)P(S_B)=AB/2^{n+m}$, which realize the multiplication of two input bitstreams S_A and S_B. The counter accumulates four stochastic streams $S_{A1,B1}$~$S_{A4,B4}$, to get the full result of polynomial.

The results of SC may be inaccurate. It is caused by the following reasons. First, two input bitstreams should be unrelated, otherwise $P(S_{A,B})=P(S_A)P(S_B)$ is false. Second, the stochastic stream is essentially a result of N_F-fold Bernoulli experiment, and the number of "1" contained in the stream will satisfy the binomial distribution[22]. The length of a stochastic bit-stream increases exponentially with the desired resolution. For example, if we want to get a probability of bitstream with 8-bit resolution, we have to observe at least 2^{16} bits of stream using classic SC [18].

Some optimization methods in precision and latency based on the classical method has been presented. The method of Fig.1.(b) is the optimization for latency, which

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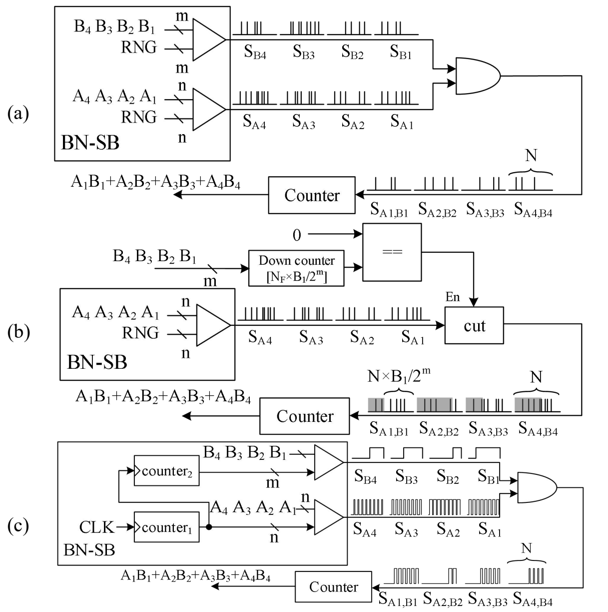

Fig. 1. Frequency encode stochastic computing, (a) Classic method[2]; (b) Binary-Interfaced Stochastic Computing(BISC)[10]; (c) Deterministic method[17]. (*A* and *B* are unsigned binary number with *n*-bit and *m*-bit respectively).

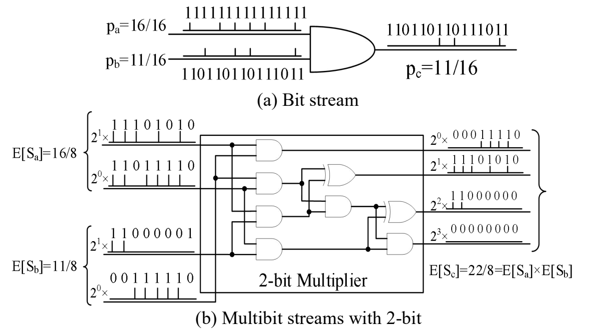

Fig. 2. Multiplication of AFE(Amplitude and Frequency Encoding) for SC.

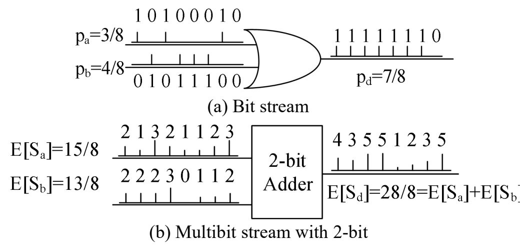

Fig. 3. Adder of AFE(Amplitude and Frequency Encoding) for SC.

can achieve 50% average latency reduction [10]; Fig.1.(c) is aimed at the optimization of precision. Using the periodicity of pseudo-random sequence, zero error calculation can be realized [17], and to get a probability of stream with (n+m)-bit resolution it only need 2^{n+m} bits.

B. Amplitude and Frequency Encoding Stochastic Computing

Previous researches usually consider that one value is represented using the probability of "1" in a stream. Using $s(1)$, $s(2),...,s(N)$ to represent bitstream. Every $s(i)$ in bit-stream is generated by a discrete random variable S with a probability distribution that $P(S{=}1){=}p$, $P(S{=}0){=}1{-}p$, where p is the probability value. According to definition of expectation in [22], we can get the expectation of S as follow,

$$E[S] = 1 \cdot p + 0 \cdot (1 - p) = p \tag{1}$$

According to (1), it can also be regarded as that the SC uses the discrete random variable S as a signal source to generate stochastic stream and $E[S]$ represents the value of stream. Since S=*1 or 0* in conventional SC , the stochastic streams are 1-bit streams.

If SC is understood as described above, we argue that the discrete random variable S should have a larger range instead of being limited to 1 or 0 only. Therefore, SC can be extend to use multibit streams instead of 1-bit streams. And the example of stochastic computing with 2-bit stream are shown as Fig.2 and Fig.3.

According to [22], there are the natures of expectation in the following (2)~(5). C is the constant and X, Y are discrete random variable,

$$E[C] = C \tag{2}$$

$$E[C{\cdot}X] = C{\cdot}E[X] \tag{3}$$

$$E[X{\cdot}Y] = E[X]E[Y], X,Y \text{ are independent} \tag{4}$$

$$E[X + Y] = E[X] + E[Y] \tag{5}$$

1) Multiplication and addition

If multiplying two multibit streams (non 1-bit stream) $s_a(i)$, $s_b(i)$ number by number, a new multibit stream $s_c(i)$ can be obtained,

$$s_c(i) = s_a(i)s_b(i) \tag{6}$$

According to (4), if $s_a(i)$, $s_b(i)$ are independent, the expectation of $s_c(i)$ is the product of $E[S_a]$ and $E[S_b]$, as shown in (7). Therefore, the multiplication of multibit streams can be realized by multiplying two numbers together at the same location of two streams and as shown in Fig.2.

$$E[S_c] = E[S_aS_b] = E[S_a]E[S_b] \tag{7}$$

If adding two multibit streams $s_a(i)$, $s_b(i)$ number by number, a new multibit stream $s_d(i)$ can be obtained,

$$s_d(i) = s_a(i) + s_b(i) \tag{8}$$

According to (5), the expectation of $s_d(i)$ is equal to $E[S_a]$+$E[S_b]$, as shown in (9). Therefore, the addition of multibit stream can be realized by adding two numbers together at the same location of two streams and as shown in Fig.3.

$$E[S_d] = E[S_a + S_b] = E[S_a] + E[S_b] \tag{9}$$

Using Fig.2(b) to explain details. The binary numbers 16 and 11 should firstly be scaled into [0, 2^2-1] to satisfied the representation range of stream with 2-bit. Therefore, we encode 2-bit streams with $E[S_a]$=16/8 and $E[S_b]$=11/8 to represent 16 and 11. Then, the 5-bit multiplication will be realized by 2-bit multiplier in Fig.3(a), and the addition computing be also realized by the multibit stream with 2-bit in Fig.3(b).

Through this new SC method for multibit stream, the new SC architecture can trade-off on latency and hardware occupation which is different from conventional SC. Based on the method proposed, the information is encoded both on amplitude and frequency of stochastic stream.

2) BN-SB converter

When encoding binary number x, $x{=}y \cdot p{=}y \cdot z/2^n$, $y \in Z \cap [0, 2^m\text{-}1]$, $z \in Z \cap [0, 2^n\text{-}1]$, $p \in [0, 1]$. The BN-SB circuit is shown in Fig.4. The bit stream $f(i)$ with p is generated by

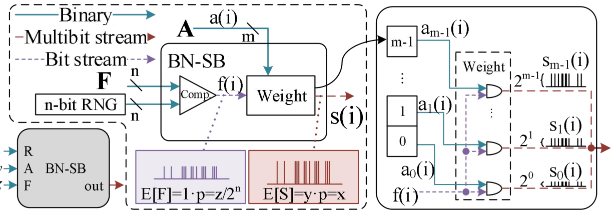

Fig. 4. The Circuit of BN-SB for AFE stream.

RNG and comparator, which is the same as BN-SB circuit for conventional SC.

$$E[F] = p = z/2^n \tag{10}$$

On the other hand, if making *a(i)=y*, *i*=1,2,...,*N*, the expectation of *A* is as follow,

$$E[A] = y \tag{11}$$

Finally, *s(i)* is obtained by multiplying *a*(*i*) and *f*(*i*) together, as shown in the right side of Fig.4. $a_0(i), a_1(i), \ldots, a_{m-1}(i)$ indicates the *m* bits of *a*(*i*). The *m* bits of *s*(*i*) and *y* has the same representation as *a*(*i*).

$$s(i) = \sum_{j=0}^{m-1} 2^j s_j(i) = \sum_{j=0}^{m-1} [2^j a_j(i) \cdot f(i)] \tag{12}$$

The values expressed by *s*(*i*) is determined by the expectation of *s*(*i*). Since *A* is a constant, we can get the expectation according to (3), as shown in (13). Therefore, *s*(*i*) is the multibit stream wanted, and the circuit achieves the purpose of converting the real-valued *x* to multibit stream.

$$E[S] = \left(\sum_{j=0}^{m-1} 2^j y_j\right) E[F] = y \cdot E[F] = y \cdot p = x \tag{13}$$

Bit streams represent different information by the different frequency of “1” pulses, while multibit streams use not only the frequency of pulse but also the amplitude of pulse to represent different information. So we refer to this new SC method with multibit streams as AFE (amplitude and frequency encoding), and refer to the multibit stream as AFE stream.

AFE has a less requirement for stream length *N*. Because *y*(*m*-bit) is encoded on the amplitude, encoding the frequency with *n*-bit resolution($N=2^m$) is enough to represent *x*(*n*+*m* bit).

3) SB-BN converter

Similar to conventional SC, the strategy of recovering values from AFE stream is still to estimate the expectation by computing the average of a long enough AFE stream, so we use accumulators to achieve the function of SB-BN.

$$mean[s(i)] = \sum_{i=1}^{N} s(i)/N \approx E[S] \tag{14}$$

To avoid division operation, *N* takes the integer power of 2, which can achieve division by shifting.

4) A special multiplier for quadratic polynomial

As the BN-SB circuit completes stream encoding, it also naturally completes a multiplication operation. Therefore, when realizing the first multiplication of polynomials, BN-SB can be used as a multiplier which uses two binary inputs to output the AFE stream. The advantage of this special usage is more prominent when operating quadratic polynomial, because the quadratic polynomial only has one multiplication in every monomial.

When dealing with the quadratic polynomial shown in Fig. 5, $x^2+y^2-2x-5y$, *x* and *y* are *m*-bit number. Binary operation will require four *m*-bit multipliers and three 2*m*-bit adders, while the AFE SC requires only four BN-SB converter and three *m*-bit adders. Therefore, the hardware occupation is reduced, since the BN-SN for AFE SC has a close level of

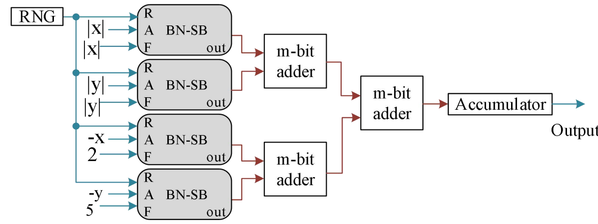

Fig. 5. BN-SB circuit for the quadratic polynomial fitting: $z=x^2+y^2-2x-5y$.

hardware occupation than conventional SC as shown in Table II.

III.ERROR DISCUSSION FOR AFE SC

A.Probability distribution of AFE stream

As previously mentioned, *s*(*i*) are generated by discrete random variables, *S*. The range of *S* can be adjusted according to use, if *S* is *m*-bit, $S \in Z \cap [0, L]$, $L=2^m-1$. The probability distribution of *S* is $P(S=0)=p_0$, $P(S=1)=p_1$, ..., $P(S=L)=p_L$.

$$E[S] = p_0 \cdot 0 + p_1 \cdot 1 + \ldots + p_L \cdot L \tag{15}$$

According to the encoding method in section II.B(2), there are two binary number $x_a=y_ap_a$ and $x_b=y_bp_b$. If the AFE streams representing x_a and x_b are generated by the BN-SB circuit, the probability distribution of S_a is $P(S_a=0)=1-p_a$, $P(S_a=y_a)=p_a$, the probability distribution of S_b is $P(S_b=0)=1-p_b$, $P(S_b=y_b)=p_b$, the expectation and of S_a and S_b are,

$$E[S_a] = (1-p_a) \cdot 0 + y_a p_a = y_a p_a = x_a \tag{16}$$

$$E[S_b] = (1-p_b) \cdot 0 + y_b p_b = y_b p_b = x_b \tag{17}$$

If $s_c(i)$ is the multiplication of $s_a(i)$ and $s_b(i)$, $s_a(i)$ and $s_b(i)$ are independent, the probability distribution of S_c can be obtained, $P(S_b=0)=1-p_ap_b$, $P(S_b=y_ay_b)=p_ap_b$.

$$E[S_c] = (1-p_ap_b) \cdot 0 + (y_ay_b)(p_ap_b) = x_ax_b \tag{18}$$

If $s_d(i)$ is the addition of $s_a(i)$ and $s_b(i)$, the probability distribution of S_d can be obtained, $P(S_d=0)=(1-p_a)(1-p_b)$, $P(S_d=y_a)=p_a(1-p_b)$, $P(S_d=y_b)=p_b(1-p_a)$, $P(S_d=y_a+y_b)=p_ap_b$.

$$\begin{aligned} E[S_d] &= (1-p_a)(1-p_b) \cdot 0 + p_a(1-p_b)y_a \\ &\quad + p_b(1-p_a)y_b + p_ap_b(y_a+y_b) \\ &= p_ay_a + p_ay_a = x_ax_b \end{aligned} \tag{19}$$

Observing the above computation, the stream output from BN-SB(Fig.3) can only be 0 or y. After the multiply operation, S_c still takes two values, 0 or y_ay_b. While after the add operation, S_d will take four values, the probability distribution of S_d becomes more complicated. If a polynomial operation is performed, the probability distribution and expectation can also be obtained according to the above process. Its probability distribution will be much more complicated caused by add operations in polynomial.

B.Precision experiment for AFE SC

As mentioned in Section I, because of the existing of progressive precision[1], precision can easily change by choosing different stream length, which is one of the advantages of SC. Due to the hardware implementation of the true RNG is more complex, the pseudo-random sequence is usually used in SC. When using pseudo-random sequence, obtaining PP through statistic becomes a better

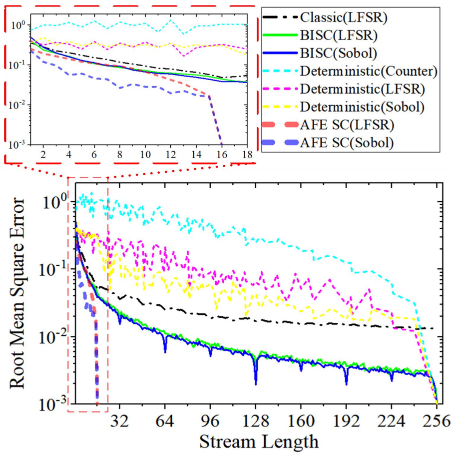

Fig. 6. The progressive precision of Classic[1], BISC[10], Deterministic[17] and AFE SC for 5-bit multiplication of two input, using different kind of RNG

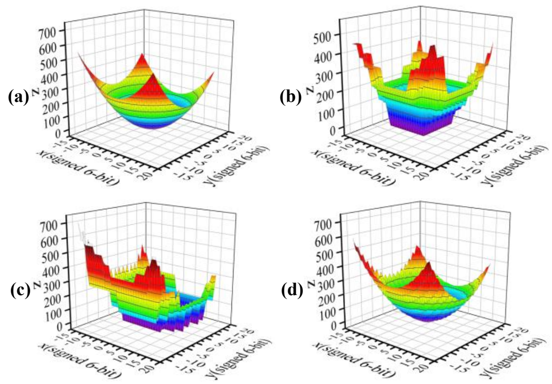

Fig. 7. Quadratic polynomial fitting: $z=x^2+y^2-2x-5y$, x,y is signed 6-bit binary and stream length N=16. (a) Binary, (b) BISC[10], (c) Deterministic [17], (d) AFE.

TABLE I
THE LATENCY OF DIFFERENT STOCHASTIC METHODS

Method			Classic[2]	AFE SC	
RNG			LFSR	LFSR	Sobol
Bit width	4 bit	RMSE	0.035	0.017	0.016
		Stream Length	2^{4+4}	15	15
	8 bit	RMSE	0.011	0.011	0.011
		Stream Length	23,104	200	30
	10 bit	RMSE	0.012	0.012	0.011
		Stream Length	38,416	440	30

RNG has the same bit width as the input data.

strategy, which will be used in following test.

Taking binary computation as the benchmark, we statistically treat the deviation in multiplication between SC and binary. RMSE (root mean square error) is calculated as the index to measure the precision (20). $SC(A,B,N)$ is the result of stochastic computing for $A\times B$ with stream length N, $P(A,B)$ is the theoretical value for $A\times B$, where A is n-bit and B is m-bit. When $RMSE(N)$=0, the result of SC is completely consistent with the binary calculations, which means a zero-error calculation is achieved.

$$RMSE(N)=\sqrt{\frac{\sum_{A=0}^{2^n-1}\sum_{B=0}^{2^m-1}[SC(A,B,N)-P(A,B)]^2}{2^{n+m}}} \tag{20}$$

Figure 6 shows the PP(progressive precision) of four methods with different RNGs when performing 4-bit multiplication(n=m=4 in Fig.1). Among the three methods with bitstream, BISC has the best progressive precision(PP). While AFE SC has a speed increasing of 2^m times more than the BISC to achieve zero error calculation. As for the progressive precision, AFE SC also has a better performance than other method.

TABLE II
THE AREA EFFICIENCY OF DIFFERENT STOCHASTIC METHODS

Method	RNG Area		Computing Core Area		Latency	Area efficiency* ($\times10^{-5}$)
	LUT	FF	LUT	FF		
Classic[2]	14	20	2	20	38,416	1.3
AFE SC	9	10	10	20	30	333
Binary	-	-	115	73	1	869.6

* $(\text{LUT}\times\text{latency})^{-1}$
In area efficiency, logic resource(LUT) only calculate the computation block, because RNG can be shared when operating quadratic polynomial.

It should be noted that the AFE SC in the test is using BN-SB circuit as multiplier. Because the period of RNG is equal to 2^n. every number in [0, 2^n-1] will appear once in the random of RNG. When N=2^n=16, the frequency encoding of bitstream $f(i)$ in Fig.4 is zero error. Therefore, AFE-SC can reach zero error calculation. Further, a quadratic polynomial fitting was performed using conventional SC and AFE SC(Fig. 5), and the results are shown in Fig. 7. Under the same stream length, AFE SC has a visible advantage.

Table I shows the latency of AFE and classic SC to reach the same precision (RMSE≈0.01). As mentioned in Section II.A deterministic[17] needs 2^{n+m}-bit to represent (n+m)-bit resolution, so the stream length N in the test is also no more than 2^{n+m}.

We implement the circuit for SC methods with 10-bit data by FPGA as shown in Table II. Latency represent the stream length required to achieve a similar precision (RMSE≈0.01) when input data is 10 bit, $(\text{LUT}\times\text{latency})^{-1}$ is counted to measure the area efficiency of each method. The area efficiency of BN-SB as multiplier is 256 times that of classic SC method with 10-bit input data.

IV. Conclusion

This paper presents a new encoding method, AFE for SC. The method proposed use the expectation of AFE streams to achieve the stochastic computing. Compared with conventional stochastic bitstream, AFE proposed realize low-latency and low-area occupation instead of the long latency of classic method. We proof the rationality of computing rules for AFE stream and design the basic circuit for AFE SC. Especially for quadratic polynomial operations, it only requires using the BN-SB circuits of AFE without additional multiplier circuits. In the experiments for precision, using BN-SB as multiplier can reduce the latency by a factor of $1/2^m$ to reach zero error calculation, and its area efficiency is 256 times that of classic SC method with 10-bit input data.

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